

10.5

13. Instrument Precision A precision instrument is guaranteed to read accurately to within 2 units. A sample of four instrument readings on the same object yielded the measurements 353, 351, 351, and 355.

- Test the null hypothesis that $\sigma = .7$ against the alternative $\sigma > .7$. Use $\alpha = .05$.
- Find a 90% confidence interval for the population variance.

$$\bar{X} = 352.5, S^2 = 3.6667, n = 4$$

$$a. H_0: \sigma^2 = \sigma_0^2 = 0.7^2, H_a: \sigma^2 > \sigma_0^2 = 0.7^2$$

$$\alpha = 0.05$$

$$\chi^2_{STAT} = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2_3$$

$$\chi^{2*} = \frac{(4-1)3.6667^2}{0.7^2} \approx 22.449$$

$$\chi^2_{3;0.05} = 7.815$$

$$\text{rejection region: } \chi^2 > 7.815$$

$$\chi^{2*} \approx 22.449 > 7.815 \Rightarrow \text{reject } H_0$$

b.

$$90\% \text{ CI } \sigma^2: \left(\frac{(n-1)S^2}{\chi^2_{3;0.05}}, \frac{(n-1)S^2}{\chi^2_{3;0.95}} \right) = (1.4676, 31.2637)$$